Considerable radioactivity is induced in the sodium in the primary cooling circuit of a liquid metal cooled fast breeder reactor (LMFBR): $^{23}Na(n,\gamma)\,^{24}Na$, $^{23}Na(n,p)\,^{23}Ne$, etc. ^{24}Na has a 15 h $t_{½}$ and emits energetic γ's. The primary cooling loop must therefore be well shielded. Its activity is a nuisance only in case of repair work in the primary system, requiring considerable waiting time before the loop can be approached. The sodium dissolves many of the corrosion and fission products eventually released. To remove these, the primary loop is provided with cold trap purification systems.

Other coolants currently discussed, e.g. in Europe, is molten lead and helium. For all coolants there are advantages and disadvantages. For example lead is excellent for accident management since a core melt will dissolve in the lead and thus disarm the situation leaving a large, heavily contaminated lead block. Also some of the more troublesome volatile fission products such as iodine will react with the hot lead and be trapped preventing their release to the environment. A considerable disadvantage with using lead as a coolant is the increased corrosion in pumps and pipes. Sometimes bismuth is added to the lead in order to lower the melting point and make the environment less aggressive. This will, however, produce polonium during reactor operation. Any release of gas from a gas-cooled reactor will contain some radioactive iodine formed from tramp uranium and leaking fuel. Thus special venting systems and filters will be needed.

20.4. REACTOR WASTE

In the "ideal" nuclear reactor all fission products and actinides produced are contained in the fuel elements. As mentioned in §19.17, there are four processes in a water cooled reactor through which radioactivity leaves the reactor vessel; in all cases the carrier of activity is the coolant:

(i) induced radioactivity in the cooling medium (and moderator if separate);
(ii) corrosion products containing induced activities (mainly from materials in the feed water);
(iii) leaked fission products and actinides from faulty fuel elements;
(iv) fission products of actinides deposited on surfaces in the core ("tramp uranium").

We order the discussion of reactor wastes according to the phase which contains the radioactivity (gas, liquid or solid) and end by a brief discussion of waste from decommissioning of nuclear reactors.

20.4.1. Gaseous wastes

The intense flux of fast and thermal neutrons induces several radioactivities in H_2O: $^{2}H(n,\gamma)^{3}H$, $^{16}O(n,p)^{16}N$, $^{18}O(n,\gamma)^{19}O$, and $^{18}O(n,p)^{18}F$. Considerable activities of ^{13}N are produced through the reactions $^{13}C(n,p)^{13}N$ and $^{14}N(n,2n)^{13}N$ with carbon in the

steel, and CO_2 and nitrogen in the water. ^{14}C is produced through reactions with ^{13}C, ^{14}N (mainly dissolved to 10–60 ppm in the fuel), and ^{16}O; 400–800 GBq ^{14}C is produced per GW_e-year, mainly contained in the fuel. Activation products from B, Li and Ar occur when the cooling water contains these atoms due to the reactions $^{10}B(n,2\alpha)^3H$, $^{6}Li(n,\alpha)^3H$, and $^{40}Ar(n,\gamma)^{41}Ar$. In a PWR the tritium content may be 40–400 GBq m^{-3} because of the boric acid and lithium content, while 0.04 GBq m^{-3} is more typical for a BWR. Though most of these activities are short lived many emit energetic γ-rays in their decay. In a BWR the active gases are transported with the steam to the turbines, where a considerable fraction of the decays occur, thus raising the background in this part of the power station to such a level that personnel cannot be allowed permanently in the turbine hall during operation. Also some radioactivity may be released through leaking gland seals; however, this can be very much reduced by adding a little inactive steam from a small electrically heated boiler to the high pressure side of the seals.

Other activities are also found in the steam: ^{24}Na, activated corrosion products like ^{60}Co, and fission products like ^{91}Sr, ^{99m}Tc, and ^{137}Cs. These activities are many powers of ten lower than for the gases.

To take care of gaseous products transported with the steam, efficient drainage and suction (air ejector) systems are installed at the condenser side of the turbines. Here most of the activity is swept out and passes through a hydrogen-oxygen recombiner and a delay system (delay times of ~30 min are used in older and ~30 h in modern BWRs) before the rest of the gases are caught in filters, consisting of absorbing material (e.g. charcoal). Because some of the activity may leak out into the reactor and turbine buildings, the air flow is from areas with low risk to areas with higher risk for contamination. The building ventilation system also contains filters before the air is vented through the stack. Figure 20.6 shows the principle of the off gas system of a modern BWR. By operating the system at reduced pressure leakage to the environment can be avoided. When the charcoal absorbers become saturated they can be heated and the adsorbed gas driven off back into the turbine condenser, thus causing a recirculation which multiplies the available decay time.

The primary release of gaseous activities in a PWR is at the de-aerator after the heat exchanger in the primary loop. The hydrogen-oxygen recombiner is also located here. The gases are compressed and stored for up to 2 weeks before they (via filters) are vented through the stack; these volumes are much smaller than for a BWR, whose off-gas also contains air leaked into the turbine on the condenser side.

If fission products are released by faulty fuel elements, the gaseous ones appear most rapidly in the cooling water: ^{133}Xe, ^{135}Xe, ^{85}Kr, ^{133}I, etc. The noble gases can be delayed in sand-filled hold-up tanks (permitting shortlived activity to decay) and the remaining activity retained in charcoal filters. Iodine can be removed by various special techniques. As mentioned above and indicated in Figure 20.6, many BWR plants cyclically

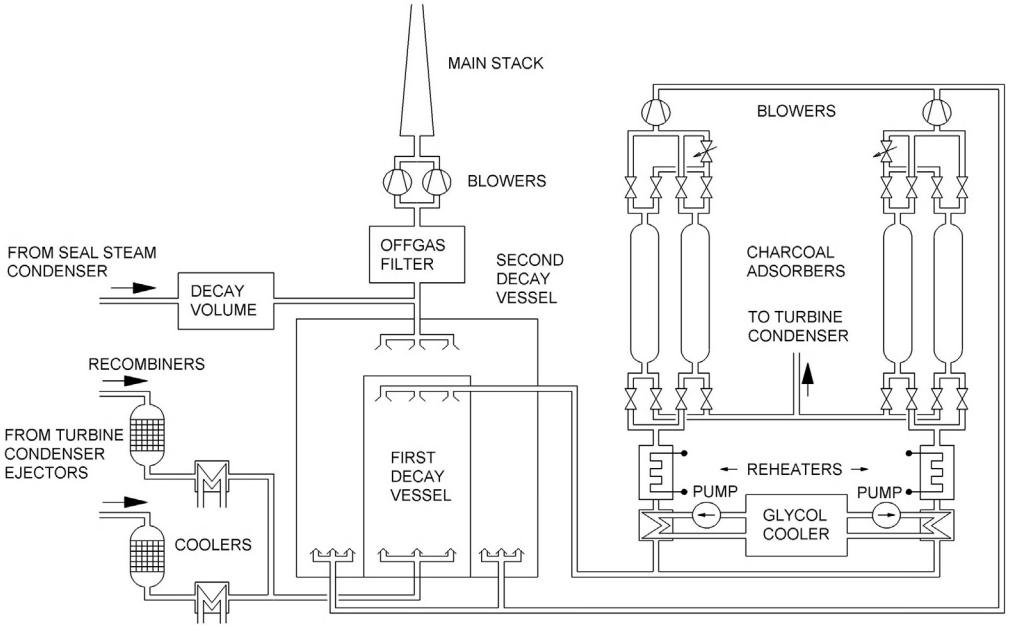


Figure 20.6 BWR ventilation and off gas treatment system (ABB Atom).

regenerate the charcoal filters feeding the released activity back into the coolant system. Some actual release values are given in Table 19.7, see also Ch. 22.

The presence of radioactive noble gases in the off-gas is monitored continuously. Abrupt increases occur when fuel pins are penetrated. By considering the different half-lives and activities it is often possible to make a rough differentiation between tramp uranium, old leaks, pinholes and large new leaks in the fuel.

20.4.2. Liquid wastes

Corrosion and fission products appear in dissolved ionic form and "precipitated" in the crud, depending on the chemistry and water conditions. Most of the corrosion products giving rise to induced activity enter with the feed water. The dominating activated corrosion products are ^{51}Cr, ^{54}Mn, ^{59}Fe, ^{58}Co, ^{60}Co, ^{65}Zn, and ^{124}Sb, and the dominating fission products are ^{3}H, ^{131}I, ^{134}Cs, and ^{137}Cs. Other fission products and actinides are released in minor amounts depending on the kind and size of fuel element leak. These products are continually removed by cleaning circuits. The aqueous chemistry, and related corrosion and waste problems, of operating reactors is one of the main areas for (nuclear) chemists working in the nuclear power industry.

In a PWR a continuous liquid stream is withdrawn from the coolant in the primary circuit, on the back side of a main heat exchanger. After cooling in another heat

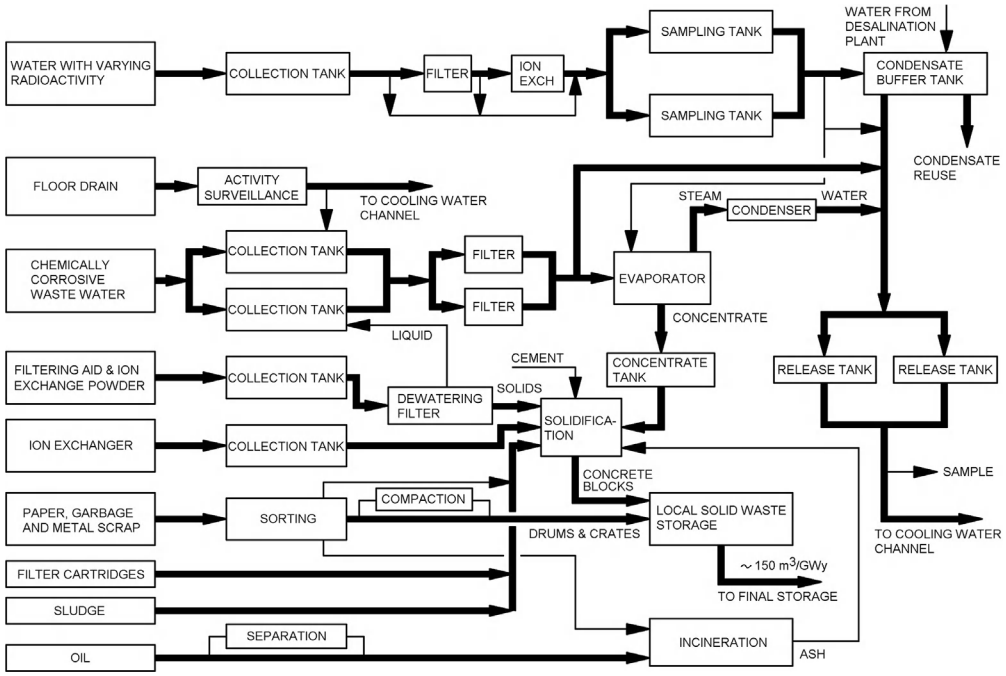


Figure 20.7 Typical waste management system for a nuclear power station, but many variations exist.

exchanger, the water is allowed to pass a filter (e.g. of the pre-coat type, 25 μm particle size) and a demineralizer which removes ionic species as well as all particulates. The liquid waste streams are detailed in Figure 20.7, which is typical for both PWRs and BWRs. The system is designed to permit segregation of waste streams according to characteristics and type of treatment. The largest liquid waste source from a PWR is the steam generator blow-down, while the largest activities originate at the clean-up systems (condensate demineralizer and filter back-washes); the "demineralizer" may be an inorganic substance (e.g. kieselguhr) or an organic ion exchanger (among the most efficient ones is the Powdex system, which consists of very finely ground, ~400 mesh, organic cation and anion exchangers). The boric acid adjustment system of some PWRs may be considered as part of the cleaning circuit; it is discussed in §19.13.1.

A BWR has three liquid purification circuits: one after the condenser for the feed water, one with withdrawal of a small stream from the reactor vessel, and one for the water in the fuel storage pool. The condensate purification system contains similar demineralizers as for the PWR. The reactor water purification system consists of a cooler followed by a mixed bed ion exchanger; also other substances have been used

(magnetite, kieselguhr, etc.). A typical capacity figure for the purification system is $\sim 100\ m^3\ h^{-1}$.

The radioactivity becomes very high in the purification systems. The organic ion-exchange resins used are limited to absorbed doses of $\sim 10^6$ Gy. The filters are back-washed and resins changed through remotely controlled systems, when measurements show that too much activity is leaking through, the pressure drop becomes too large, or water quality drops. At present resins are not regenerated.

As an additional safety, when filters and demineralizers show insufficient purification of circulating water and of water for release to the environment, an evaporator system may be incorporated. Such a system is normally a part of the PWR boric acid control circuit.

A nuclear power plant contains numerous liquid streams, not all indicated in Figure 20.7, and small controlled amounts of radioactivity are released from the plant. Typical releases are given in Table 19.7 and Chapter 22; these are actual measured values, as pointed out in §19.16 normally constituting a small fraction of the legally permitted releases.

20.4.3. Solid wastes

Solid radioactive wastes are produced from many systems and purification circuits of the reactor station, see Figure 20.7. Many contaminated items can be decontaminated by proper treatment, saving both money and waste storage space. Low level waste can also sometimes be "declassified" and disposed of as normal, inactive, waste or scrap. Combustible solids may be reduced in volume through incineration. Several techniques are used for fixation of wastes in such a way that they can be safely stored with a minimum of surveillance. These techniques all have in common an enclosure of the activity in an insoluble material (normally concrete or bitumen) and in blocks of sizes and activities small enough to be handled with a fork lift.

The waste amounts are typically $\sim 500\ m^3$ ILW and $\sim 5000\ m^3$ LLW annually from a 1000 MW_e power station, but vary considerably between different stations depending on purification and concentration techniques used. The following annual averages are thought to be fairly representative for most LWRs: spent ion-exchange resins $\sim 30\ m^3$ ($\sim 50\%$ water), evaporator residues $0-20\ m^3$ ($\sim 80\%$ water), filter (condenser) concentrates $\sim 10\ m^3$, other wastes from primary reactor loop ~ 5 t. The activity in the resins is typically 2–7 TBq ^{60}Co, ~ 0.5 TBq ^{137}Cs, and ~ 0.05 TBq ^{90}Sr, and in evaporator and filter concentrates it is $<1/10$ of the resin amount.

Figure 20.8 shows waste solidification and fixation with cement, using prefabricated concrete boxes. This technique is useful both for contaminated or activated equipment and for resins, sludges, etc. The technique, however, increases the waste volume considerably, by a factor 4 to 40 times the untreated waste.

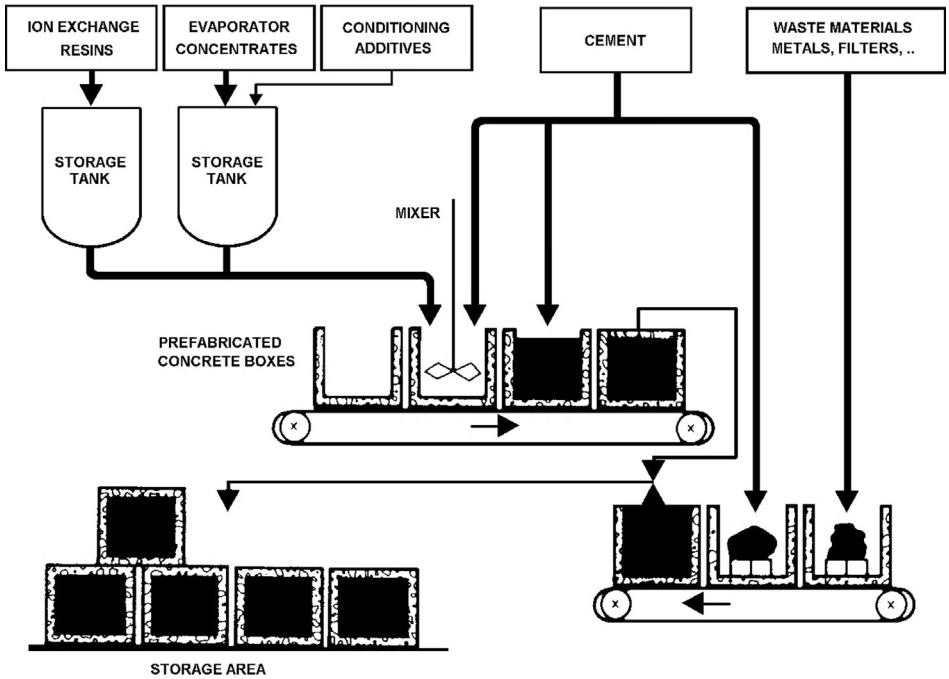


Figure 20.8 Waste solidification in concrete blocks.

Concrete is a cheap fire and corrosion resistant material. However, active species, especially easily soluble ions such as caesium, can be leached from it by water. Addition of plastic binders to the concrete in order to improve its properties has been suggested. In the continuous bitumen extruder process for semi-liquid wastes all water is directly eliminated, considerably reducing the waste volume. The bitumen mixture is placed in steel drums of standard size (150—200 l). When additional shielding is desired, the filled drums are placed into disposable or reusable sleeves of concrete, iron, or lead. Such a sleeve of 12 cm lead weighs 7 t, and reduces the surface dose rate by a factor of $\sim 10^3$. A typical unshielded bitumen drum may have a surface dose rate of 1 Sv h^{-1}, necessitating remote handling.

Comparing cement and bitumen, bitumen produces smaller waste volumes (often 5 times less than concrete) with lower initial leach rates. The disadvantage of bitumen is its in-flammability and lower radiation resistance; a fire could release large activities of fairly long-lived nuclides. Further, bitumen in wet areas has poor ageing properties, slowly deteriorating. However, for low and medium active solid wastes, bitumen is presently the preferred fixation material at many power stations. About 300 bitumen drums (200 l) with concentrated waste are produced annually per 1300 MW_e reactor in Germany;

about 50 of these contain medium active waste and have outer concrete shielding (outer volume ~400 l).

If the waste does not contain any fission products, a storage time of 50 y would be sufficient to allow all activity (mainly ^{60}Co) to decay to negligible amounts; storage at plant site would be feasible. If the amount of ^{90}Sr, ^{137}Cs, or actinides is high, much longer storage times are required, and a special storage site will be required. If possible, it is desirable to separate these products from other waste; as an alternative the isolated ^{90}Sr, ^{137}Cs, and actinides could then be added to the high level waste of reprocessing plants.

The cementation and bituminisation techniques are also used for intermediate level wastes arising in other parts of the nuclear fuel cycle, e.g. reprocessing. The long-term storage of the solidified waste is described in Chapter 21 and some of its possible future impact on the environment in Chapter 22.

20.4.4. Decommissioning nuclear reactors

Most of the waste from the dismantling of decommissioned nuclear reactors is non-radioactive. Also, large volumes can be decontaminated and "declassified" which permits them to be treated as normal industrial waste. Of the remaining waste, most contains low levels of relatively short-lived nuclides similar to wastes from normal reactor operation. Hence, it can be treated and disposed of in the same manner, e.g. put in bitumen or concrete, and stored in repositories for LLW and ILW (§21.9). The volumes may be rather large, however. Some internal parts of the reactor have relatively high activity. The induced activity is limited to part of the pressure vessel (closest to the core) and some core components.

The inner surface of the pressure vessel is normally covered by highly radioactive deposits. These deposits can be dissolved and removed by a repeated treatment with suitable chemicals. As an example, after more than 20 years of operation the pressure vessel of the BWR reactor Oskarshamn 1, Sweden, was decontaminated to such an extent that workers could enter the empty pressure vessel to perform necessary modifications and tests with only a cylindrical radiation shield placed inside the central part of the vessel wall to protect against induced radioactivity in the wall. Upper and lower parts of the pressure vessel are normally fairly well shielded from the neutron flux and only contain a low level of induced radioactivity. For chemical decontamination, i.e. dissolution of deposits, weak acids will dissolve hydroxides and complex formers (such as chelating agents) bind metal ions in soluble complexes. However, in some countries (e.g. Sweden) authorities forbid the inclusion of strong complex formers in the reactor waste. Therefore, in the case of Oskarshamn 1, oxalic acid was used; after dissolution of the deposits and fixing the metal ions on ion-exchange resins oxalic acid was destroyed by ultraviolet light yielding CO_2 and H_2O. Thus, a pressure vessel of a decommissioned reactor could, after chemical decontamination, be cut into smaller sections, some of

which would only be slightly radioactive. The resulting scrap can either be encapsulated in concrete and put in a repository for normal reactor waste (Swedish plans), or melted to be used as raw material for new reactor parts where the remaining level of activity is acceptable (France).

20.5. SAFE OPERATION OF NUCLEAR REACTORS

The major risks from reactor operation stem from operator mistakes (<10%) or failure of critical hardware (>90%). The risk from human errors is reduced by recurrent training of personnel, especially persons working in the control room, using mockups and simulators. Thereby personnel in critical functions train their behaviour in normal situations, but also in situations which seldom or never occur during routine operation of the reactor. Also exchange of experience and the results from analysis of actions violating or threatening to violate safety instructions help to reduce the risk for human errors. Risks from hardware failures or malfunctions are minimized by a continuous work aiming to identify risk dominating sequences and critical hardware. Once identified, such sequences should be modified, procedures changed, equipment replaced or duplicated in order to reduce the overall risk until it reaches an acceptable level. It is important to remember that simple duplication will not reduce the risk from common mode failures very much.

Figure 20.9 shows an example of an *event tree* starting with an *initiating event* to the left and branching each time an action takes place. The upward branch corresponds to a success of the action specified at the top, whereas the downward branch represents a failure. Each branch is normally the compounded effect of actions in a sub-tree describing the function and interaction of affected hardware. The figures at each downward branch are estimated probabilities for a failure of the action at the top. The line representing the outcome of an event is then followed to the right; either to a final state or to another branch. To the extreme right is the final state with its estimated probability. For example, the most important risk in Fig. 20.9 arises from failure to restore AC power within 20 minutes; probability $(1/25) \times 10^{-6} = 4 \times 10^{-8}$ per reactor year. This was the main reason for the Fukoshima accident where the auxiliary diesel generators were drowned by the tsunami. For independent outcomes of events in a branch (e.g. probability of failure of too many of the available safety valves together with failing operation of the blow-down pipes), the probability is the product of the probabilities at the branches taken in the sub-tree. In case of common mode failures, the compounded probability is the same as the probability of a single event of that type. The probabilistic event tree is an example of the type of tools used to identify the current risk dominating event chain. When carried out meticulously, operator training, experience feedback and risk analysis help to continually improve the safety of reactor operation.